1. Introduction to Finite Element Method

1.1 What is FEM?

- The Finite Element Method (FEM) is a numerical technique used to find approximate solutions of boundary value problems in engineering.
- The whole structure (continuum) is divided into a finite number of small pieces called finite elements, connected at points called nodes.
- Each element has a simple behaviour described by a displacement (shape) function; the behaviour of the whole body is obtained by assembling all elements.
- Result: a large set of simultaneous algebraic equations $[K]\{u\} = \{F\}$ which is solved by computer.

1.2 Brief history

- 1943 Courant introduced the idea using triangular elements for torsion problems.
- 1956 Turner, Clough, Martin and Topp used FEM for aircraft structural analysis (stiffness method).
- 1960 Clough coined the term "Finite Element Method".
- Later extended to heat transfer, fluid flow, electromagnetics, vibration, etc.

1.3 General steps in FEM

- Step 1: Discretise the structure (divide into elements and nodes).
- Step 2: Select a suitable displacement function (shape function).
- Step 3: Derive the strain-displacement and stress-strain relations.
- Step 4: Derive the element stiffness matrix and load vector.
- Step 5: Assemble element equations to get the global equation $[K]\{u\}=\{F\}$.
- Step 6: Apply boundary conditions (supports and loads).

- Step 7: Solve for the nodal displacements.
- Step 8: Calculate element strains and stresses (post-processing).

1.4 Applications of FEM

- Structural analysis: beams, trusses, frames, plates, shells, bridges, dams.
- Mechanical: stress concentration, machine parts, automobile and aircraft bodies.
- Thermal analysis: conduction and convection in solids.
- Fluid flow, seepage through soil, electromagnetic fields, vibration and buckling.

1.5 Advantages of FEM

- Can handle complex geometry, irregular boundaries and any shape of body.
- Can handle different material properties, anisotropic and non-homogeneous materials.
- Different types of boundary conditions and loads can be applied easily.
- The same general program can be used for many different problems.
- Element size can be varied: fine mesh where stress changes rapidly, coarse mesh elsewhere.
- Dynamic, non-linear and time dependent problems can also be solved.
- Saves cost of prototypes; the design can be modified quickly on the computer.

1.6 Disadvantages of FEM

- Requires a computer and good, costly software; large memory and computing time.
- The result is only approximate, not exact; accuracy depends on the mesh.
- Skill is needed to prepare the model; wrong inputs give wrong results (garbage in, garbage out).

- Large amount of input data and output data to be checked.
- Stress values at nodes may show errors (discontinuity of stress between elements).
- Not economical for very simple problems where a closed form solution is available.

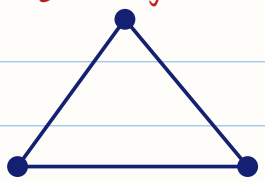
FEM = approximate method; Exact solution (theory of elasticity) is not possible for complex shapes.

1-D

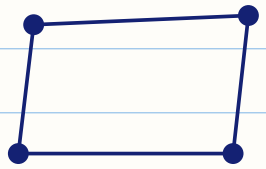


line / bar / beam

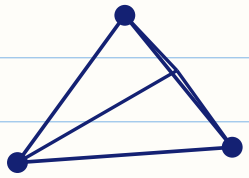
2-D triangle



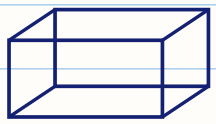
2-D quad



3-D tetrahedron



3-D hexahedron (brick)



2. Basic Equations in Elasticity

2.1 Stress at a point

- Stress = internal resisting force per unit area. Normal stress σ (perpendicular to the face) and shear stress τ (parallel to the face).
- In 3-D, six independent stress components act at a point:

$$\{\sigma\} = \{\sigma_x \; \sigma_y \; \sigma_z \; \tau_{xy} \; \tau_{yz} \; \tau_{zx}\}^T$$

- Moment equilibrium of an element gives $\tau_{xy} = \tau_{yx}$, $\tau_{yz} = \tau_{zy}$, $\tau_{zx} = \tau_{xz}$.

2.2 Strain at a point

- Strain = change in dimension per unit original dimension. Normal strain ε and shear strain γ.

$$\{\varepsilon\} = \{\varepsilon_x\ \varepsilon_y\ \varepsilon_z\ \gamma_{xy}\ \gamma_{yz}\ \gamma_{zx}\}^T$$

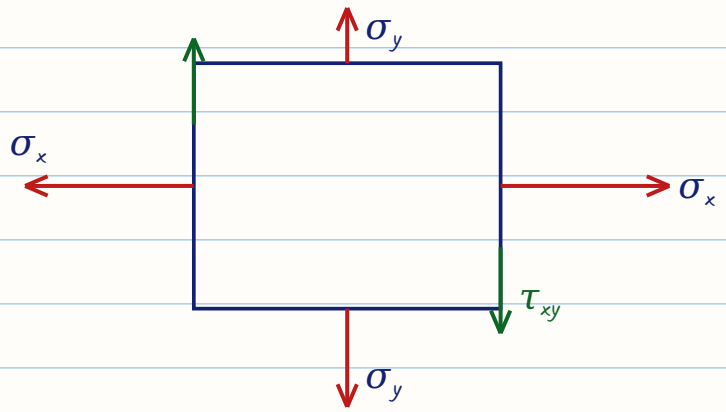
- If u, v, w are the displacements in the x, y, z directions, then the strain-displacement relations are:

$$\varepsilon_x = \partial u/\partial x,\ \varepsilon_y = \partial v/\partial y,\ \varepsilon_z = \partial w/\partial z$$

$$\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$$

$$\gamma_{yz} = \partial v/\partial z + \partial w/\partial y$$

$$\gamma_{zx} = \partial w/\partial x + \partial u/\partial z$$



2-D stress element

$$\sigma = \{\sigma_x\ \sigma_y\ \tau_{xy}\}^T$$

$$\varepsilon = \{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T$$

$\tau_{xy} = \tau_{yx}$ (complementary shear)

2.3 Equilibrium equations (3-D)

- For an element in equilibrium (body forces X, Y, Z per unit volume):

$$\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y + \partial\tau_{xz}/\partial z + X = 0$$

$$\partial\tau_{xy}/\partial x + \partial\sigma_y/\partial y + \partial\tau_{yz}/\partial z + Y = 0$$

$$\partial\tau_{xz}/\partial x + \partial\tau_{yz}/\partial y + \partial\sigma_z/\partial z + Z = 0$$

2.4 Stress – strain relation (Hooke's law)

- For a linear elastic, isotropic material, with E = Young's modulus, ν = Poisson's ratio, G = shear modulus:

$$\varepsilon_x = [\sigma_x - \nu(\sigma_y + \sigma_z)] / E$$

$$\varepsilon_y = [\sigma_y - \nu(\sigma_z + \sigma_x)] / E$$

$$\varepsilon_z = [\sigma_z - \nu(\sigma_x + \sigma_y)] / E$$

$$\gamma_{xy} = \tau_{xy}/G,\ \gamma_{yz} = \tau_{yz}/G,\ \gamma_{zx} = \tau_{zx}/G,\ G = E / 2(1+\nu)$$

- In matrix form $\{\sigma\} = [D]\{\varepsilon\}$ where $[D]$ is the elasticity (material) matrix.

2.5 [D] matrix for 3-D (isotropic)

$$[D] = E(1-\nu) / [(1+\nu)(1-2\nu)] \times [\text{matrix}]$$

- The 6×6 matrix has: diagonal terms for normal stress = 1, coupling terms = $\nu/(1-\nu)$, shear diagonal terms = $(1-2\nu) / 2(1-\nu)$; all others are zero.

3-D problems need 6×6 [D]. Plane problems reduce the size to 3×3.

3. Concept of Plane Stress

3.1 Definition

- Plane stress occurs in a THIN plate loaded only in its own plane (in-plane loads), with thickness t very small compared to the other dimensions.
- Stresses in the z-direction (thickness) are zero:

$$\sigma_z = 0,\ \tau_{xz} = 0,\ \tau_{yz} = 0$$

- Non-zero stresses: σ_x, σ_y, τ_{xy}. But strain $\varepsilon_z \neq 0$ (the plate becomes thinner or thicker).
- Examples: thin plate with a hole, gear tooth (thin), deep beam, thin disc, plate under in-plane tension.

3.2 Stress-strain relation for plane stress

$$\varepsilon_x = (\sigma_x - \nu\,\sigma_y) / E$$

$$\varepsilon_y = (\sigma_y - \nu\,\sigma_x) / E$$

$$\gamma_{xy} = \tau_{xy} / G = 2(1+\nu)\,\tau_{xy} / E$$

- Solving for the stresses, $\{\sigma\} = [D]\{\varepsilon\}$:

$$\{\sigma_x\ \sigma_y\ \tau_{xy}\}^T = E/(1-\nu^2)\,[\,1\ \nu\ 0\,;\,\nu\ 1\ 0\,;\,0\ 0\ (1-\nu)/2\,]\,\{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T$$

Plane stress $[D] = E / (1-\nu^2) \times [\,1\ \nu\ 0 / \nu\ 1\ 0 / 0\ 0\ (1-\nu)/2\,]$

- Out-of-plane strain: $\varepsilon_z = -\nu\,(\sigma_x + \sigma_y) / E$

4. Concept of Plane Strain

4.1 Definition

- Plane strain occurs in a LONG body whose cross-section and loading do not change along the length (z-axis), and the ends are restrained.
- Strains in the z-direction are zero:

$$\varepsilon_z = 0,\ \gamma_{xz} = 0,\ \gamma_{yz} = 0$$

- Non-zero strains: ε_x, ε_y, γ_{xy}. But $\sigma_z \neq 0$ ($\sigma_z = \nu(\sigma_x + \sigma_y)$).
- Examples: retaining wall, long dam, tunnel, pipe under internal pressure, rolling cylinder, foundation under strip footing.

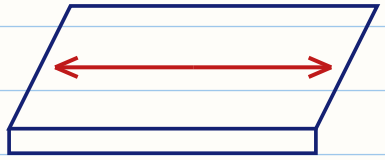
4.2 Stress-strain relation for plane strain

$$\varepsilon_x = (1+\nu)/E\,[\,(1-\nu)\sigma_x - \nu\,\sigma_y\,]$$

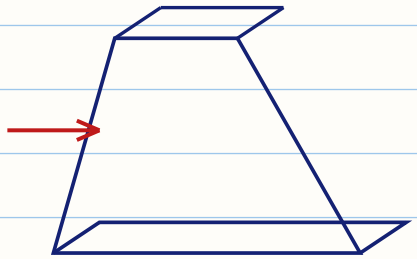
$$\varepsilon_y = (1+\nu)/E\,[\,(1-\nu)\sigma_y - \nu\,\sigma_x\,]$$

$$\{\sigma_x\ \sigma_y\ \tau_{xy}\}^T = E(1-\nu)\,/\,[(1+\nu)(1-2\nu)]\,[\,1\ \nu/(1-\nu)\ 0\,;\,\nu/(1-\nu)\ 1\ 0\,;\,0\ 0\ (1-2\nu)/2(1-\nu)\,]\,\{\varepsilon\}$$

Plane strain $[D] = E / [(1+\nu)(1-2\nu)] \times [\,1-\nu\ \nu\ 0 / \nu\ 1-\nu\ 0 / 0\ 0\ (1-2\nu)/2\,]$



Plane stress: thin plate
t small, $\sigma_z=\tau_{xz}=\tau_{yz}=0$



Plane strain: long body
$\varepsilon_z=\gamma_{xz}=\gamma_{yz}=0$

4.3 Plane stress vs plane strain

- Plane stress: $\sigma_z = 0, \varepsilon_z \neq 0$. Plane strain: $\varepsilon_z = 0, \sigma_z \neq 0$.
- Plane stress is for thin bodies; plane strain is for thick/long bodies.
- To change a plane stress [D] into plane strain [D], replace E by $E/(1-\nu^2)$ and ν by $\nu/(1-\nu)$.
- Both reduce a 3-D problem to a 2-D problem, with only three stress and three strain components.

5. Element Shapes, Nodes and Degrees of Freedom

5.1 Finite element

- A finite element is a small, simple sub-region of the structure. All elements together form the "mesh".
- Element shapes depend on the dimension of the problem:
- 1-D elements: straight or curved line (bar, truss, beam, spring) with 2 or 3 nodes.
- 2-D elements: triangle (3 or 6 nodes), quadrilateral (4, 8 or 9 nodes) – plane stress, plane strain, plates, shells.
- 3-D elements: tetrahedron (4 or 10 nodes), hexahedron/brick (8 or 20

nodes), wedge.

- Triangles fit curved and irregular boundaries well; quadrilaterals give better accuracy per element.

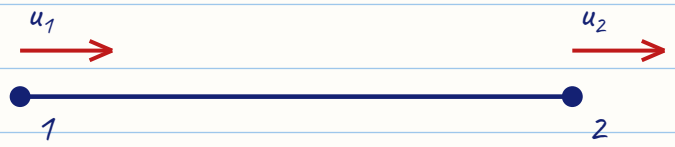
5.2 Nodes

- Nodes are the specific points of an element where the unknowns (displacements) are calculated. Elements are connected to each other only at nodes.
- Exterior (corner) nodes define the geometry. Interior / mid-side nodes are added for higher order elements.
- Nodes should be placed at points of load application, supports, change in thickness / material / geometry.

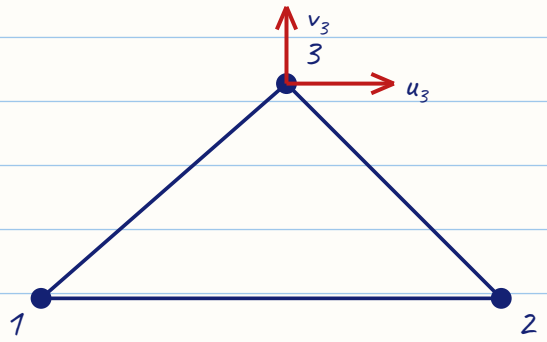
5.3 Nodal degrees of freedom (dof)

- Degrees of freedom = number of independent displacements (or rotations) of a node.
- Bar / truss (1-D): 1 dof per node (u). Plane truss: 2 dof per node (u, v).
- Beam element: 2 dof per node (deflection v and slope θ).
- 2-D plane stress / strain element: 2 dof per node (u, v).
- 3-D solid element: 3 dof per node (u, v, w). Frame / shell: up to 6 dof per node.

  Total dof of element = (number of nodes) × (dof per node)
- Example: CST has 3 nodes × 2 = 6 dof. 8-node brick has 8 × 3 = 24 dof.



1-D bar: 1 dof per node



2-D: 2 dof per node (u, v)

6. Displacement Function (Shape Function)

6.1 Concept

- The actual displacement inside an element is unknown, so we ASSUME a simple function of position, usually a polynomial. It is called the displacement function.
- The polynomial must give the displacement at every point inside the element in terms of the nodal displacements.
- Polynomial is preferred because it is easy to differentiate and integrate.

1-D bar: $u = a_1 + a_2x$ (2 constants for 2 nodes)

2-D triangle: $u = a_1 + a_2x + a_3y$, $v = a_4 + a_5x + a_6y$

2-D 4-node quad: $u = a_1 + a_2x + a_3y + a_4xy$

Number of terms in polynomial = number of dof of the element.

6.2 Writing in terms of shape functions

$$\{u\} = [N]\{q\}$$

- $\{u\}$ = displacement at any point, $[N]$ = matrix of shape functions, $\{q\}$ = nodal displacement vector.
- Properties of shape functions:
- $N_i = 1$ at node i and $N_i = 0$ at all other nodes.

- Sum of all shape functions = 1 at every point (rigid body motion is represented).
- Shape functions are continuous within the element.

6.3 Convergence requirements of displacement function

- Compatibility: displacement must be continuous across element boundaries (no gaps or overlaps).
- Completeness: the function must include rigid body motion and constant strain states.
- The polynomial should be complete up to its order, and geometrically isotropic (no preferred direction) – use Pascal's triangle.

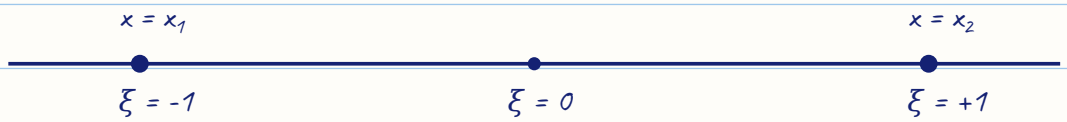
Pascal's triangle : 1 ; x y ; x^2 xy y^2 ; x^3 x^2y xy^2 y^3 ...

7. Natural Coordinates

7.1 Meaning

- A natural (local) coordinate system is attached to the element itself. Its value lies between -1 and +1 (or 0 and 1) and does not depend on the actual size or position of the element.
- Shape functions are written in natural coordinates, which makes the formulation general and suitable for numerical integration.
- Types: 1-D natural coordinate ξ, area (triangular) coordinates L_1, L_2, L_3, and volume coordinates for a tetrahedron.

7.2 1-D natural coordinate



$\xi = 2(x - x_1) / (x_2 - x_1) - 1$

$N_1 = (1-\xi)/2, \quad N_2 = (1+\xi)/2$

$x = N_1x_1 + N_2x_2,\ N_1 = (1-\xi)/2,\ N_2 = (1+\xi)/2$

- Jacobian $J = dx/d\xi = (x_2-x_1)/2 = l/2$, so $dx = (l/2)\, d\xi$.

7.3 Area coordinates for a triangle

- A point P inside a triangle 1-2-3 divides it into three small areas A_1, A_2, A_3.

  $L_1 = A_1/A,\ L_2 = A_2/A,\ L_3 = A_3/A$, A = total area

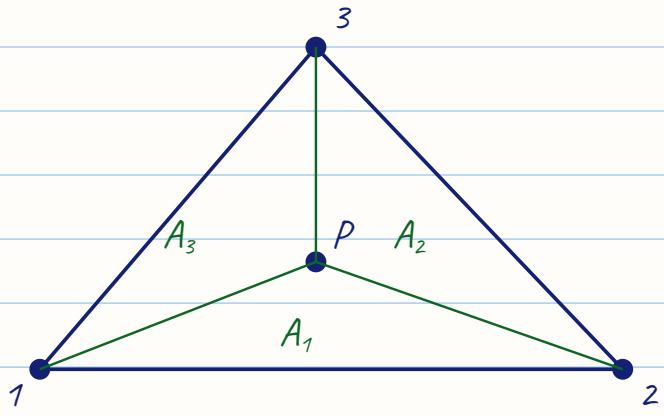
  $L_1 + L_2 + L_3 = 1$ (only two are independent)

- Linear shape functions of a triangle are equal to the area coordinates: $N_1 = L_1$, $N_2 = L_2$, $N_3 = L_3$.

  $x = L_1x_1 + L_2x_2 + L_3x_3,\ y = L_1y_1 + L_2y_2 + L_3y_3$

- Integration formula (very useful):

  $\int L_1^a L_2^b L_3^c\, dA = [a!\, b!\, c! / (a+b+c+2)!] \times 2A$



$L_i = A_i / A$

$L_1+L_2+L_3 = 1$

$L_i = 1$ at node i,

$L_i = 0$ on opposite side

8. Strain – Displacement Relations

8.1 For 1-D bar element

- With $u = N_1u_1 + N_2u_2$, the axial strain is:

  $\varepsilon = du/dx = (u_2 - u_1) / l$

  $\{\varepsilon\} = [B]\{q\}$, $[B] = [\,-1/l \;\; 1/l\,]$
- Stress $\sigma = E\,\varepsilon = E\,[B]\{q\}$.

8.2 For 2-D element

- Strains are found by differentiating the assumed displacements:

  $\{\varepsilon\} = \{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T = \{\partial u/\partial x\ \partial v/\partial y\ \partial u/\partial y + \partial v/\partial x\}^T$
- Writing $u = [N]\{q\}$:

  $\{\varepsilon\} = [B]\{q\}$
- $[B]$ is the strain-displacement matrix; its terms are derivatives of the shape functions.
- Element stress:

  $\{\sigma\} = [D]\{\varepsilon\} = [D][B]\{q\}$
- Element stiffness matrix (thickness t, volume V):

  $[k] = \int [B]^T [D] [B]\, dV = t \int [B]^T [D][B]\, dA$

For a triangle with linear shape functions, $[B]$ is constant; therefore strain and stress are constant over the element (CST).

8.3 Important formulae to remember

- Strain energy $U = \tfrac{1}{2} \int \{\varepsilon\}^T \{\sigma\}\, dV = \tfrac{1}{2} \{q\}^T [k] \{q\}$
- Potential energy $\Pi = U - W$ (W = work done by external loads)
- Principle of minimum potential energy: $\partial\Pi/\partial q = 0$ gives $[k]\{q\} = \{F\}$.

Short questions: (1) define node, (2) state any four advantages of FEM, (3)

differentiate plane stress and plane strain, (4) write the [D] matrix for plane stress, (5) what is a shape function?

1. Classification of Elements

1.1 Based on shape function order

- Simplex element: the displacement polynomial has constant and linear terms only (2-node bar, 3-node triangle, 4-node tetrahedron).
- Complex element: polynomial has higher order terms in addition to the linear terms (e.g. 6-node triangle, 8-node quad).
- Multiplex element: the boundaries are parallel to the coordinate axes and the polynomial has higher terms (e.g. rectangle with 4 nodes).

1.2 Regular and irregular elements

- Regular element: the shape is geometrically simple and symmetric – equal sides and angles (equilateral triangle, square, cube). Easy to formulate.
- Irregular element: the sides are of unequal length and the angles are arbitrary (general triangle, general quadrilateral, skewed brick). Needed to fit actual structures.
- Irregular elements are handled using natural coordinates / isoparametric mapping.
- Distorted (very skewed) elements reduce the accuracy – keep aspect ratio near 1 and angles between 45° and 135°.

1.3 2-D and 3-D elements

- 2-D: triangle (CST 3-node, LST 6-node), rectangle / quadrilateral (4, 8, 9 nodes).
- 3-D: tetrahedron (4 or 10 nodes), hexahedron or brick (8 or 20 nodes), prism / wedge (6 or 15 nodes).
- Higher order nodes lie on the mid-side of edges, giving curved boundaries and better accuracy.

2. Lagrangian and Serendipity Elements

2.1 Lagrange family

- Based on Lagrange interpolation polynomials. The element has nodes at the corners, mid-sides AND an interior node (for higher order).
- The shape function is the product of 1-D Lagrange polynomials in ξ and η:

$$N_i(\xi,\eta) = l_j(\xi) \cdot l_k(\eta)$$

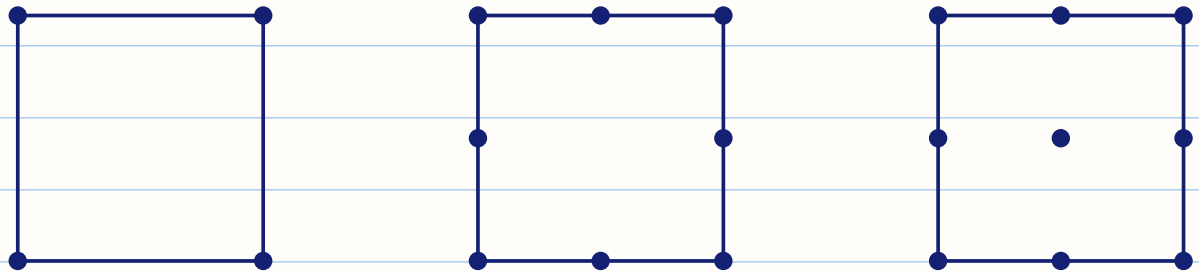
- 4-node (bilinear), 9-node (biquadratic), 16-node (bicubic) rectangles are Lagrange elements.
- Number of nodes = $(n+1)^2$ for polynomial order n.
- Disadvantage: interior nodes are extra unknowns and do not help connect neighbouring elements – they increase the size of the problem.

2.2 Serendipity family

- "Serendipity" means a lucky discovery. Nodes are only on the boundary (corners and mid-sides), NO interior node.
- Shape functions are not obtained by simple products; they are built by inspection and checked: each $N_i = 1$ at node i, 0 at others.
- 4-node, 8-node (quadratic) and 12-node (cubic) elements. Fewer nodes than Lagrange for same order, so cheaper.
- The polynomial of an 8-node serendipity element misses the x^2y^2 term:

$$u = a_1 + a_2\xi + a_3\eta + a_4\xi^2 + a_5\xi\eta + a_6\eta^2 + a_7\xi^2\eta + a_8\xi\eta^2$$

Serendipity: no interior node Lagrange: has interior node



4-node bilinear 8-node serendipity 9-node Lagrange

2.3 Comparison

- Lagrange: more nodes (9 vs 8), complete polynomial, interior node, higher cost.
- Serendipity: fewer nodes, no interior node, more economical, good for curved boundaries.
- Both give the same order of accuracy for the same polynomial order,

so serendipity is preferred in practice.

2.4 Shape functions of 4-node rectangle (Lagrange)

- With natural coordinates ξ, η from -1 to +1 and node numbering anticlockwise from lower-left:

$$N_1 = \tfrac{1}{4}(1-\xi)(1-\eta),\ N_2 = \tfrac{1}{4}(1+\xi)(1-\eta)$$

$$N_3 = \tfrac{1}{4}(1+\xi)(1+\eta),\ N_4 = \tfrac{1}{4}(1-\xi)(1+\eta)$$

$$N_1+N_2+N_3+N_4 = 1$$

2.5 Shape functions of 8-node serendipity element

- Corner nodes ($\xi_i, \eta_i = \pm 1$):

$$N_i = \tfrac{1}{4}(1+\xi\xi_i)(1+\eta\eta_i)(\xi\xi_i + \eta\eta_i - 1)$$

- Mid-side nodes with $\xi_i = 0$:

$$N_i = \tfrac{1}{2}(1-\xi^2)(1+\eta\eta_i)$$

- Mid-side nodes with $\eta_i = 0$:

$$N_i = \tfrac{1}{2}(1+\xi\xi_i)(1-\eta^2)$$

2.6 Shape functions of 9-node Lagrange element

Corner: $N_i = \tfrac{1}{4}\xi\xi_i(1+\xi\xi_i)\,\eta\eta_i(1+\eta\eta_i)$

Mid-side: $N_i = \tfrac{1}{2}(1-\xi^2)\,\eta\eta_i(1+\eta\eta_i)$ or $\tfrac{1}{2}\xi\xi_i(1+\xi\xi_i)(1-\eta^2)$

Centre: $N_9 = (1-\xi^2)(1-\eta^2)$

3. Hermite Polynomials

3.1 Concept

- In Lagrange / serendipity elements only the function value (displacement) is a nodal unknown – this gives C^0 continuity (the displacement is continuous but its slope is not).
- Hermite interpolation uses BOTH the value and its first derivative (slope) at each node, giving C^1 continuity.
- Needed for beams, plates and shells where slope must also be continuous.
- For an element with 2 nodes and 2 unknowns per node $(v, \theta) \rightarrow 4$ conditions → cubic polynomial:

$$v(x) = a_1 + a_2x + a_3x^2 + a_4x^3$$

3.2 Hermite shape functions ($s = x / l$)

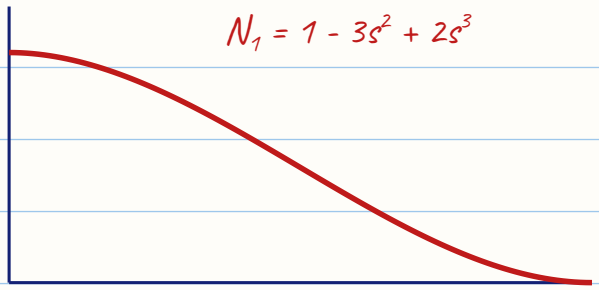
$$H_1 = 1 - 3s^2 + 2s^3$$

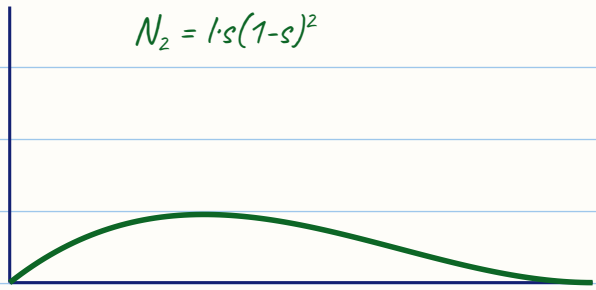
$$H_2 = l\,(s - 2s^2 + s^3)$$

$$H_3 = 3s^2 - 2s^3$$

$$H_4 = l\,(-s^2 + s^3)$$

- Conditions: $H_1(0)=1$, $H_3(l)=1$ give the displacement; $H_2'(0)=1$, $H_4'(l)=1$ give the slope.





$s = x / l$ (Hermite cubics)

4. Shape Functions up to Quadratic Formulation

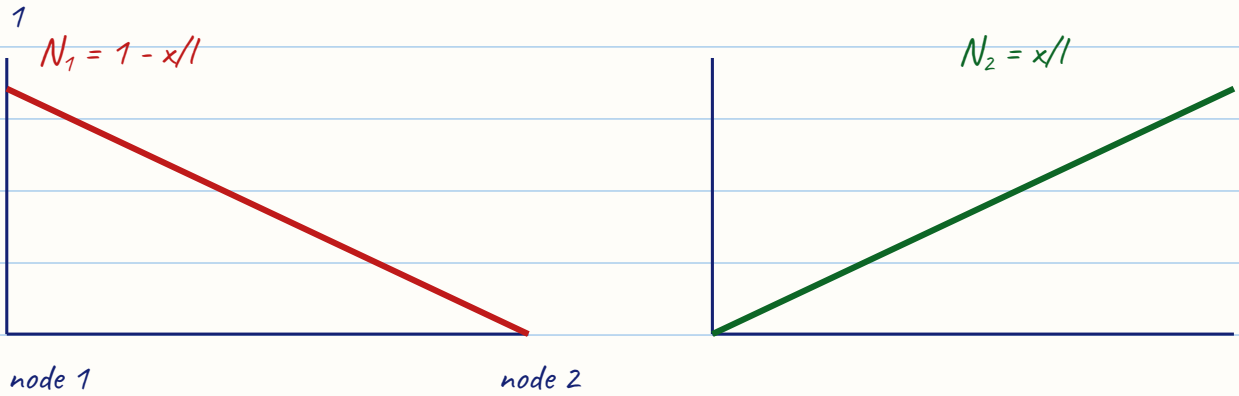
4.1 Linear 1-D element (2 nodes)

- Assume $u(x) = a_1 + a_2 x$. Apply at nodes: $u(0) = u_1$, $u(l) = u_2$.

$a_1 = u_1$, $a_2 = (u_2 - u_1) / l$

$u = (1 - x/l)\, u_1 + (x/l)\, u_2 = N_1 u_1 + N_2 u_2$

$N_1 = 1 - x/l$, $N_2 = x/l$

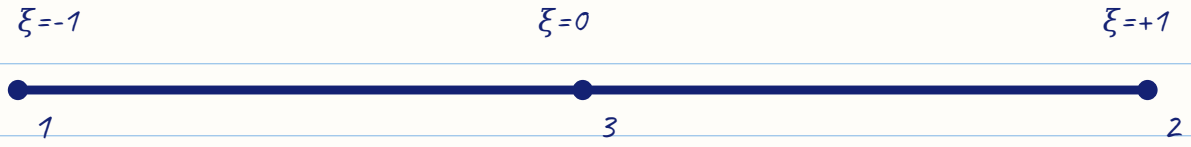


4.2 Quadratic 1-D element (3 nodes)

- Assume $u(\xi) = a_1 + a_2\xi + a_3\xi^2$ with nodes at $\xi = -1, +1$ and 0 (mid-node).

  $N_1 = \frac{1}{2}\xi(\xi - 1)$, $N_2 = \frac{1}{2}\xi(\xi + 1)$, $N_3 = 1 - \xi^2$

- Check: at $\xi = -1$, $N_1 = 1$; at $\xi = 1$, $N_2 = 1$; at $\xi = 0$, $N_3 = 1$; sum = 1.



$N_1 = \xi(\xi-1)/2$, $N_2 = \xi(\xi+1)/2$, $N_3 = 1-\xi^2$

3-node (quadratic) bar element

4.3 Linear triangle (3 nodes)

- $u = a_1 + a_2x + a_3y$. Solving with nodal values gives

  $N_i = (a_i + b_ix + c_iy) / 2A$, $i = 1, 2, 3$

  $a_1 = x_2y_3 - x_3y_2$, $b_1 = y_2 - y_3$, $c_1 = x_3 - x_2$

- (others by cyclic permutation $1 \to 2 \to 3 \to 1$) and $2A = x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)$.

4.4 Quadratic triangle (6 nodes) in area coordinates

$$N_1 = L_1(2L_1 - 1),\ N_2 = L_2(2L_2 - 1),\ N_3 = L_3(2L_3 - 1)$$

$$N_4 = 4L_1L_2,\ N_5 = 4L_2L_3,\ N_6 = 4L_3L_1$$

- Nodes 1-3 are corners, nodes 4-6 are mid-sides.

4.5 Linear tetrahedron (4 nodes)

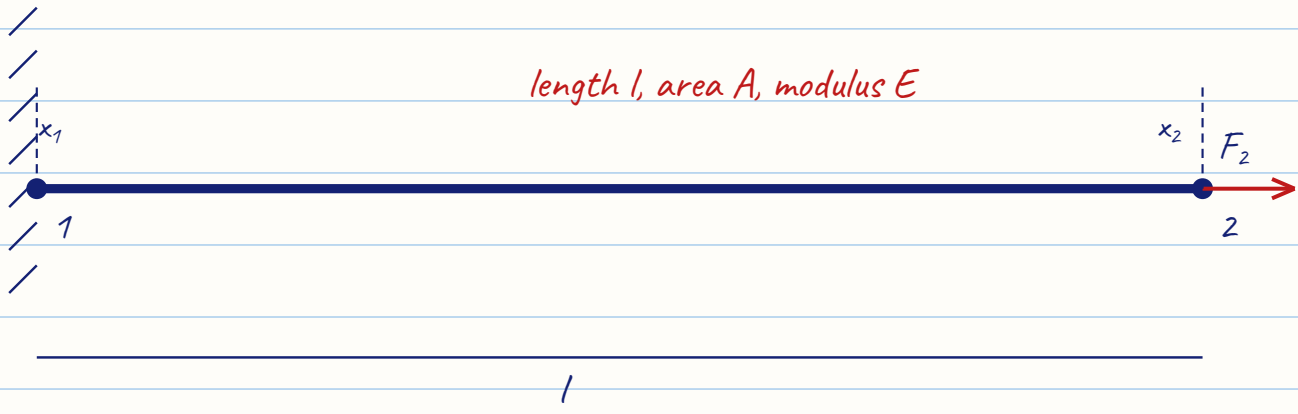
$$N_i = (a_i + b_ix + c_iy + d_iz) / 6V$$

- In volume coordinates $N_i = L_i$, with $L_1+L_2+L_3+L_4 = 1$.

5. FEA of One-Dimensional Problems – Bar Element

5.1 Bar element

- A bar (rod) carries only axial load. Each node has one dof (axial displacement u).
- Let length = l, cross-sectional area = A, Young's modulus = E.



$$u(x) = N_1u_1 + N_2u_2$$

5.2 Displacement function and shape functions

$$u = N_1 u_1 + N_2 u_2 = [N]\{q\},\ [N] = [\,1 - x/l \quad x/l\,]$$

$$\{q\} = \{u_1\ u_2\}^T$$

5.3 Strain – displacement relation

$$\varepsilon = du/dx = [\,-1/l \quad 1/l\,]\{u_1\ u_2\}^T = [B]\{q\}$$

$$[B] = (1/l)[\,-1 \quad 1\,]$$

- Strain in a linear bar element is constant.

5.4 Stress – strain relation

$$\sigma = E\varepsilon = E[B]\{q\} = (E/l)(u_2 - u_1)$$

- Axial force in the bar $P = \sigma A = (AE/l)(u_2 - u_1)$.

5.5 Stiffness matrix by direct method (from Hooke's law)

- Force at node 2: $F_2 = (AE/l)(u_2 - u_1)$; for equilibrium $F_1 = -F_2$.

$$\{F_1\ F_2\}^T = (AE/l)[\,1 \ -1\ ;\ -1 \ \ 1\,]\{u_1\ u_2\}^T$$

$$[k] = (AE/l)[\,1 \ -1\ ;\ -1 \ \ 1\,]$$

5.6 Stiffness matrix by energy method

- Strain energy of the element:

$$U = \tfrac{1}{2}\int \sigma\,\varepsilon\,A\,dx = \tfrac{1}{2}\{q\}^T\left[\int [B]^T E [B] A\,dx\right]\{q\}$$

$$[k] = \int_0^l [B]^T E [B] A\,dx = AEl[B]^T[B]$$

$$[k] = (AE/l)[\,1 \ -1\ ;\ -1 \ \ 1\,]$$

- The matrix is symmetric, singular (determinant = 0) because rigid

body motion is not yet prevented, and has diagonal terms always positive.

5.7 Load vectors

- Body force f per unit volume (e.g. self weight, ρg):

  $\{f_b\} = A f l / 2 \{1\ 1\}^T$

- Traction / distributed load T per unit length:

  $\{f_T\} = T l / 2 \{1\ 1\}^T$

- Point loads are applied directly at the nodes.

5.8 Temperature effects

- Thermal strain $\varepsilon_0 = \alpha \Delta T$. Equivalent nodal force:

  $\{f_{th}\} = A E \alpha \Delta T \{-1\ 1\}^T$

5.9 Bar of variable cross-section

- Use the mean area $A = (A_1 + A_2) / 2$ for a tapered bar, or divide into several elements of constant area.

  $[k] = (E (A_1+A_2) / 2l) [\,1\ -1\ ;\ -1\ 1\,]$

5.10 Quadratic (3-node) bar element

- Nodes: 1 at $\xi = -1$, 2 at $\xi = +1$, 3 at $\xi = 0$; and $dx = (l/2)\, d\xi$.

  $[B] = (2/l) [\,\xi - ½\ \xi + ½\ -2\xi\,]$

  $[k] = (AE / 3l) [\,7\ 1\ -8\ ;\ 1\ 7\ -8\ ;\ -8\ -8\ 16\,]$

6. Assembly of Bar Elements – Procedure

- Step 1: Divide the bar into elements; number the nodes and elements.
- Step 2: Write [k] of each element in terms of local dof.
- Step 3: Assemble into the global stiffness matrix [K] by adding terms at common dof.

  $$[K] = \Sigma\, [k^{(e)}]$$
- Step 4: Global equation $[K]\{U\} = \{F\}$.
- Step 5: Apply boundary conditions ($u = 0$ at the fixed support) by the elimination method or the penalty method.
- Step 6: Solve for nodal displacements U.
- Step 7: Find element stresses $\sigma = E\,[B]\{q\}$ and support reactions $R = K\,U - F$.

6.1 Illustration – two element stepped bar

- Two elements (1-2) and (2-3) with $(AE/l)_1 = k_1$, $(AE/l)_2 = k_2$.

  $$[K] = [\,k_1 \; -k_1 \; 0 \;;\; -k_1 \; k_1+k_2 \; -k_2 \;;\; 0 \; -k_2 \; k_2\,]$$
- Fixed at node 1: $U_1 = 0$, load P at node 3 gives

  $$U_2 = P/k_1,\; U_3 = P/k_1 + P/k_2$$
- Stresses: $\sigma_1 = E_1(U_2 - U_1)/l_1$, $\sigma_2 = E_2(U_3 - U_2)/l_2$.

6.2 Elimination and penalty methods

- Elimination: strike off the row and column of the fixed dof, solve the remaining equations.
- Penalty approach: add a very large number C ($C \approx 10^4 \times \max K_{ij}$) to

the diagonal term of the fixed dof, and C·a to the force vector (a = prescribed displacement).

$$(K_{ii} + C)\, U_i = F_i + C\, a$$

7. Typical Problem (Solved Example)

- Problem: A steel bar of length 400 mm is fixed at one end and has two segments: segment 1, $A = 600\ mm^2$, $l = 200$ mm; segment 2, $A = 400\ mm^2$, $l = 200$ mm. $E = 200$ GPa. A load of 20 kN acts at the free end. Find displacements and stresses.

$k_1 = A_1E/l_1 = 600 \times 200000 / 200 = 6 \times 10^5$ N/mm

$k_2 = A_2E/l_2 = 400 \times 200000 / 200 = 4 \times 10^5$ N/mm

After applying $U_1 = 0$: $[k_1+k_2\ -k_2\ ;\ -k_2\ k_2\ ]\ \{U_2\ U_3\} = \{0\ 20000\}$

$U_2 = 20000 / 6\times10^5 = 0.0333$ mm, $U_3 = 0.0333 + 20000/4\times10^5 = 0.0833$ mm

$\sigma_1 = E\,(U_2 - 0)/l = 200000 \times 0.0333/200 = 33.3\ N/mm^2$

$\sigma_2 = 200000 \times (0.0833 - 0.0333)/200 = 50\ N/mm^2$

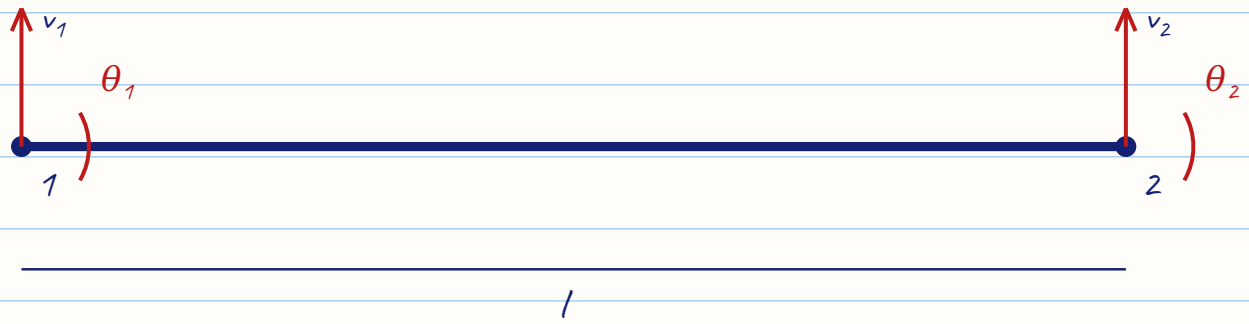
- Check: $\sigma_1 = P/A_1 = 20000/600 = 33.3$ □. $\sigma_2 = 20000/400 = 50$ □.

Exam tip: derive [k] for the bar (both methods), shape functions, assembly and one numerical problem are the usual long questions.

1. Introduction to Beam Elements

1.1 Beam element

- A beam is a long slender member that carries transverse loads and resists them by bending.
- Theory used: Euler-Bernoulli beam theory – plane sections remain plane and perpendicular to the neutral axis, shear deformation is ignored.
- Each node of the beam element has 2 degrees of freedom: transverse deflection v and slope (rotation) $\theta = dv/dx$.
- Nodal forces correspond to shear force F and bending moment M.



EI = constant; 2 dof per node (v, θ) → 4 dof per element

1.2 Assumptions

- Material is linearly elastic, homogeneous and isotropic.
- Cross-section is constant along the element; EI is constant.
- Deflections are small; slope dv/dx is small.
- Loads act in the plane of symmetry; loads are applied only at nodes

(equivalent nodal loads for UDL).

1.3 Basic beam relations

$M = EI\, d^2v/dx^2$, curvature $\kappa = d^2v/dx^2$

Shear force $F = dM/dx$, Load $w = dF/dx$

Bending strain $\varepsilon = -y\, d^2v/dx^2$, $\sigma = E\varepsilon = -E\, y\, d^2v/dx^2$

Governing equation: $EI\, d^4v/dx^4 = w(x)$

2. Displacement Function for the Beam Element

2.1 Cubic polynomial

- Four nodal dof $(v_1, \theta_1, v_2, \theta_2)$ → four constants → cubic displacement:

$$v(x) = a_1 + a_2x + a_3x^2 + a_4x^3$$

$$\theta(x) = dv/dx = a_2 + 2a_3x + 3a_4x^2$$

- Boundary conditions at the two nodes:

at $x = 0$: $v = v_1$, $\theta = \theta_1$

at $x = l$: $v = v_2$, $\theta = \theta_2$

2.2 Solving for constants

$$a_1 = v_1,\ a_2 = \theta_1$$

$$a_3 = -3(v_1 - v_2)/l^2 - (2\theta_1 + \theta_2)/l$$

$$a_4 = 2(v_1 - v_2)/l^3 + (\theta_1 + \theta_2)/l^2$$

3. Shape Functions of the Beam Element

- Substitute the constants and rearrange: $v = [N]\{q\}$, $\{q\} = \{v_1\ \theta_1\ v_2\ \theta_2\}^T$.

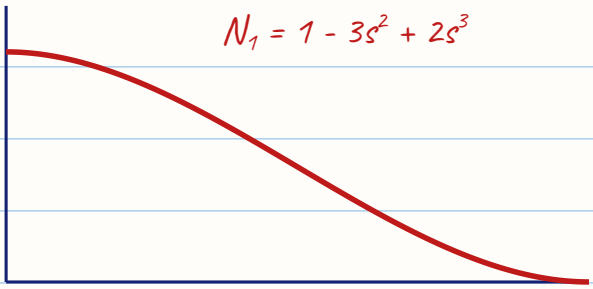
$$N_1 = 1 - 3x^2/l^2 + 2x^3/l^3$$

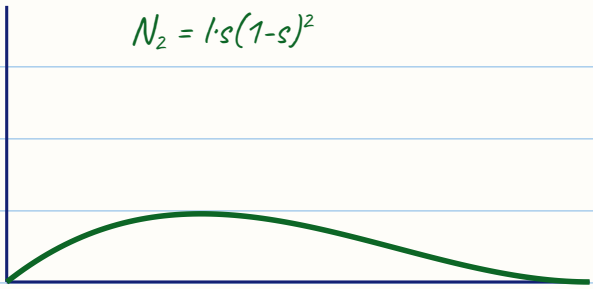
$$N_2 = x - 2x^2/l + x^3/l^2$$

$$N_3 = 3x^2/l^2 - 2x^3/l^3$$

$$N_4 = -x^2/l + x^3/l^2$$

- These are the Hermite shape functions (C^1 continuity).
- Properties: $N_1(0) = 1$, $N_3(l) = 1$; $dN_2/dx = 1$ at $x = 0$; $dN_4/dx = 1$ at $x = l$; $N_1 + N_3 = 1$.





$s = x / l$ (Hermite cubics)

4. Stiffness Matrix of the Beam Element

4.1 Strain–displacement matrix

- Curvature $d^2v/dx^2 = [B]\{q\}$ where $[B] = d^2[N]/dx^2$:

$$[B] = [\,-6/l^2 + 12x/l^3,\ -4/l + 6x/l^2,\ 6/l^2 - 12x/l^3,\ -2/l + 6x/l^2\,]$$

4.2 Strain energy approach

- Strain energy of the beam element:

$$U = \tfrac{1}{2} \int_0^l EI \left(d^2v/dx^2\right)^2 dx = \tfrac{1}{2} \{q\}^T [k] \{q\}$$

$$[k] = \int_0^l EI\, [B]^T [B]\, dx$$

4.3 Result

$$[k] = (EI/l^3) \begin{bmatrix} 12 & 6l & -12 & 6l \\ 6l & 4l^2 & -6l & 2l^2 \\ -12 & -6l & 12 & -6l \\ 6l & 2l^2 & -6l & 4l^2 \end{bmatrix}$$

- Order of dof: $v_1, \theta_1, v_2, \theta_2$. Corresponding forces: F_1, M_1, F_2, M_2.

$$\{F_1\ M_1\ F_2\ M_2\}^T = [k] \{v_1\ \theta_1\ v_2\ \theta_2\}^T$$

- Properties: symmetric, singular (rigid body motion possible), each column is a set of equilibrium forces for a unit displacement.

4.4 Physical meaning of the terms

- Column 1: unit deflection at node 1 ($v_1=1$, others 0) needs forces $12EI/l^3$, $6EI/l^2$, $-12EI/l^3$, $6EI/l^2$.
- Term $4EI/l$ = moment needed for unit rotation at one node with the other end fixed; $2EI/l$ = carry-over moment.
- Term $12EI/l^3$ = force required for unit deflection with no rotation at either end.

Remember: EI/l^3 is common factor; the matrix is 4×4 for one beam element.

5. Equivalent Nodal Loads (Load Vector)

5.1 Consistent load vector

- A distributed load $w(x)$ is replaced by equivalent nodal forces and moments:

$$\{f\} = \int_0^l [N]^T w(x)\, dx$$

5.2 Uniformly distributed load w over the whole element

$$\{f\} = \{\, wl/2 \;\; wl^2/12 \;\; wl/2 \;\; -wl^2/12 \,\}^T$$

5.3 Point load P at distance a from node 1 (b = l - a)

$$\{f\} = \{\, Pb^2(l+2a)/l^3 \;\; Pab^2/l^2 \;\; Pa^2(l+2b)/l^3 \;\; -Pa^2b/l^2 \,\}^T$$

- For a central load P: $\{f\} = \{\, P/2 \;\; Pl/8 \;\; P/2 \;\; -Pl/8 \,\}^T$.

5.4 Linearly varying (triangular) load, zero at node 1 and w_0 at node 2

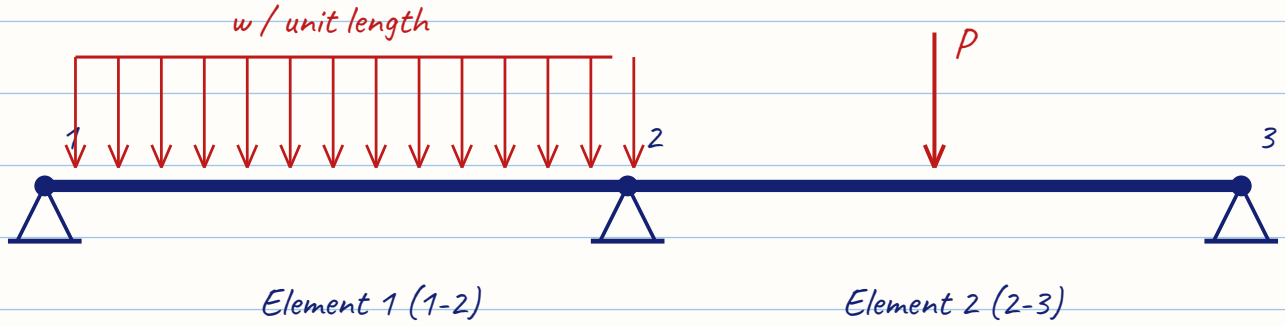
$$\{f\} = \{\, 3w_0l/20 \;\; w_0l^2/30 \;\; 7w_0l/20 \;\; -w_0l^2/20 \,\}^T$$

The fixed-end reactions of the same load have the opposite sign – they are the "fixed end forces".

6. Analysis of Continuous Beams

6.1 Continuous beam

- A beam resting on more than two supports. It is statically indeterminate; FEM treats it directly without extra compatibility equations.
- Nodes are placed at supports, at load points and at changes of section / material.



Continuous beam: elements joined at the support nodes

6.2 Procedure

- Step 1: Number the nodes (supports, load points) and elements. Each node has 2 dof (v, θ); total dof = 2 × number of nodes.
- Step 2: Find the stiffness matrix [k] of each element (4×4) using its own EI and l.
- Step 3: Find the equivalent nodal load vector of each element and combine it with applied point loads/moments.
- Step 4: Assemble the global stiffness matrix [K] by adding overlapping terms at common nodes (dof of shared node are the same for both elements).
- Step 5: Write $[K]\{U\} = \{F\}$.
- Step 6: Apply boundary conditions: deflection v = 0 at every support; slope θ = 0 at a fixed end. Eliminate those rows and columns.
- Step 7: Solve for the remaining unknown dof (slopes at supports, deflections at free points).
- Step 8: Find the support reactions and the element end forces:

$$\{f^{(e)}\} = [k^{(e)}]\{q^{(e)}\} - \{f_{equivalent}\}$$

- Step 9: Draw the shear force and bending moment diagrams.

6.3 Assembly for two elements

- Two elements (1-2) and (2-3), nodes 1, 2, 3; dof order $v_1\ \theta_1\ v_2\ \theta_2\ v_3\ \theta_3$.

  [K] is 6×6. Terms of the shared node 2 are added:

  $K_{33} = (12EI/l^3)_1 + (12EI/l^3)_2$, $K_{34} = (-6EI/l^2)_1 + (6EI/l^2)_2$

  $K_{44} = (4EI/l)_1 + (4EI/l)_2$

7. Solved Example – Propped / Two-span Beam

- Problem: A continuous beam ABC, simply supported at A and C and at B. Span AB = BC = L, EI constant, and carries a UDL of w per unit length on both spans. Find the slopes at the supports.
- Elements: (1) A–B, (2) B–C. dof: $v_A = v_B = v_C = 0$ (supports), slopes θ_A, θ_B, θ_C unknown.

  Reduced equations (rows/cols of θ only):

  $(EI/L)\ [\,4\ 2\ 0\,;\,2\ 8\ 2\,;\,0\ 2\ 4\,]\ \{\theta_A\ \theta_B\ \theta_C\}^T = \{\,wL^2/12\ 0\ -wL^2/12\,\}^T$

- By symmetry $\theta_B = 0$ (middle support):

  $(EI/L)(4\theta_A + 2\theta_B) = wL^2/12 \rightarrow \theta_A = wL^3/48EI$

  $\theta_C = -wL^3/48EI$, $\theta_B = 0$

- Support moment at B from element 1 ($\theta_B = 0$):

  $M_B = (2EI/L)\,\theta_A + wL^2/12 = wL^2/24 + wL^2/12 = wL^2/8$ (hogging)

- This matches the result of the three-moment equation.

Always check: sum of reactions = total load, and the moment at the end simple supports = 0.

8. Beam with Axial Load – Frame Element (Short Note)

- When axial deformation is also present, a plane frame element has 3 dof per node (u, v, θ) → 6×6 stiffness matrix: the bar terms AE/l are combined with the beam terms EI/l^3.

  $[k] = [\,AE/l\ 0\ 0\ -AE/l\ 0\ 0\ ;\ 0\ 12EI/l^3\ 6EI/l^2\ 0\ -12EI/l^3\ 6EI/l^2\ ;\ \ldots\ ]$

- A transformation matrix [T] is used to convert local to global axes for inclined members.

9. Derivation of [k] Term by Term (Detailed Steps)

- Start from the [B] matrix. Write the products term by term and integrate from 0 to l:

  $k_{11} = EI\int_0^l (-6/l^2 + 12x/l^3)^2\,dx = 12EI/l^3$

  $k_{12} = EI\int_0^l (-6/l^2 + 12x/l^3)(-4/l + 6x/l^2)\,dx = 6EI/l^2$

  $k_{22} = EI\int_0^l (-4/l + 6x/l^2)^2\,dx = 4EI/l$

  $k_{24} = EI\int_0^l (-4/l + 6x/l^2)(-2/l + 6x/l^2)\,dx = 2EI/l$

- Other terms follow from symmetry $k_{ij} = k_{ji}$ and from equilibrium of the element (sum of forces = 0, sum of moments = 0).
- Check by equilibrium: for column 1 the forces are 12, 6l, -12, 6l (× EI/l^3); sum of vertical forces = 12 - 12 = 0, and moment about node

$2 = 6l + 6l - 12\,l = 0$ □.

10. Solved Example – Cantilever Beam (Single Element)

- Problem: Cantilever of length $L = 2$ m, $EI = 2 \times 10^4$ kN·m^2, tip load $P = 10$ kN. Find the tip deflection and slope.
- One element, node 1 fixed ($v_1 = \theta_1 = 0$), node 2 free, load $F_2 = -10$ kN, $M_2 = 0$.

  Reduced equation: $(EI/L^3)\,[12 \;\; -6L ; -6L \;\; 4L^2]\,\{v_2 \;\; \theta_2\}^T = \{-P \;\; 0\}^T$

  From 2nd row: $\theta_2 = 6L\,v_2 / 4L^2 = 1.5\,v_2 / L$

  From 1st row: $(EI/L^3)(12\,v_2 - 9\,v_2) = -P \rightarrow v_2 = -P\,L^3 / 3EI$

  $v_2 = -10 \times 8 / (3 \times 20000) = -1.333 \times 10^{-3}$ m (downward)

  $\theta_2 = 1.5\,v_2/L = -PL^2/2EI = -1.0 \times 10^{-3}$ rad
- Both values match the exact beam theory results $\delta = PL^3/3EI$ and $\theta = PL^2/2EI$, because the cubic shape function is exact for a beam with end loads only.
- Reactions: $\{F_1 \; M_1\} = [k]\{q\}$ gives $F_1 = +10$ kN, $M_1 = +20$ kN·m □.

11. Solved Example – Fixed Beam with Central Load

- Problem: Beam of span L, both ends fixed, central point load P, EI constant. Use two elements with a node at the middle.
- Unknown dof only at the middle node: v_2, θ_2. By symmetry $\theta_2 = 0$.

  Equation (middle deflection): $2 \times 12EI/L_e^3 \times v_2 = -P$, $L_e = L/2$

  $(24\,EI \times 8 / L^3)\,v_2 = -P \rightarrow v_2 = -PL^3 / 192EI$
- This agrees with the exact result for the fixed beam.

- End moments from element 1: $M_1 = (EI/L_e^2) \times 6 \times v_2 = -PL/8$ (hogging).

Fixed-end moment for central load = PL/8; deflection under load = $PL^3/192EI$.

12. Solved Example – Simply Supported Beam with UDL

- Problem: A beam of length L carries UDL w. Use one element, supports at both ends, $v_1 = v_2 = 0$.

  Reduced equation (rows 2 and 4): $(EI/L)\,[\,4\ 2\,;\,2\ 4\,]\,\{\theta_1\ \theta_2\}^T = \{\,wL^2/12\ -wL^2/12\,\}^T$

- Add the two equations and use symmetry $\theta_2 = -\theta_1$:

  $(EI/L)(4 - 2)\,\theta_1 = wL^2/12 \rightarrow \theta_1 = wL^3/24EI$

- This is the exact slope at a simply supported end under UDL.
- Mid-span deflection from the shape functions ($N_2 = L/8$, $N_4 = -L/8$ at $x = L/2$):

  $v(L/2) = (L/8)\theta_1 - (L/8)\theta_2 = L\,\theta_1/4 = wL^4/96EI = 0.0104\ wL^4/EI$

- The exact value is $5wL^4/384EI = 0.0130\ wL^4/EI$, so one element is about 20% low at mid-span; more elements give better accuracy.

Note: the nodal values are exact for beams, but the deflection between nodes depends on the mesh – use more elements for loads between nodes.

13. Stress Recovery in Beam Elements

- Bending stress in an element at distance y from the neutral axis:

  $\sigma = -E\, y\, d^2v/dx^2 = -E\, y\, [B]\{q\}$

- [B] varies linearly along the element, so moment varies linearly: $M(x) = EI\,[B]\{q\}$.
- The nodal end forces give the shear force and bending moment at the nodes:

  $V = F_1$ (at node 1), $M = M_1$

  $\sigma_{max} = M\, y_{max} / I = M / Z$ (Z = section modulus)

14. Beam on Elastic Foundation (Short Note)

- When the beam rests on an elastic support (soil), a foundation stiffness matrix is added to the beam [k]:

  $[k_f] = (k_s\, b\, l / 420)\, [\,156\ \ 22l\ \ 54\ \ -13l\ ;\ 22l\ \ 4l^2\ \ 13l\ \ -3l^2\ ;\ 54\ \ 13l\ \ 156\ \ -22l\ ;\ -13l\ \ -3l^2\ \ -22l\ \ 4l^2\,]$

- k_s = subgrade modulus, b = width of beam. Total $[k] = [k_{beam}] + [k_f]$.

15. Quick Revision Points

- Beam element: 2 nodes, 4 dof, cubic displacement, C^1 continuity.
- [k] is 4×4, symmetric and singular; each node has v and θ.
- Equivalent nodal loads for UDL: $wl/2$ and $wl^2/12$.
- Continuous beam: assemble element matrices, apply $v = 0$ at supports, solve for slopes.
- Exact results are obtained for nodal values with cubic elements when

loads are at nodes.

Important questions: (1) derive shape functions and stiffness matrix for a beam element, (2) analyse a continuous beam (2 or 3 span) with UDL and point load, (3) write the equivalent nodal loads for UDL.

1. Two-Dimensional Problems in FEM

1.1 Introduction

- Many real problems (thin plates, dams, retaining walls, gears, tunnels) can be idealised as 2-D problems – plane stress or plane strain.
- Each node has 2 dof: displacements u (x-direction) and v (y-direction).
- The simplest 2-D element is the triangle, which fits any boundary shape, so it is the most popular one.
- Two types of triangular elements: CST (Constant Strain Triangle, 3 nodes) and LST (Linear Strain Triangle, 6 nodes).

1.2 General procedure for 2-D analysis

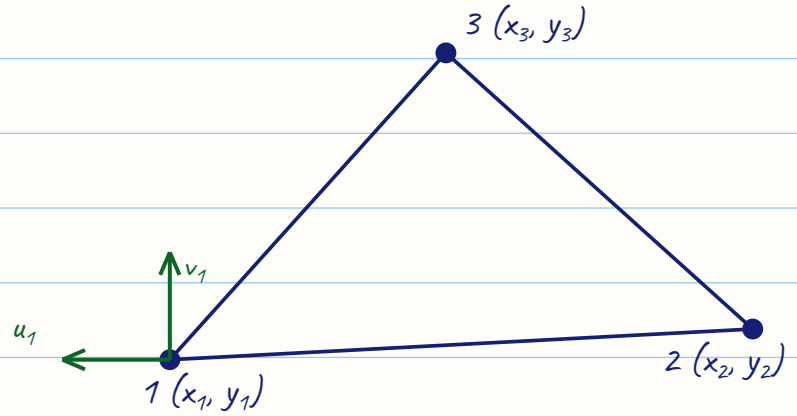
- Divide the body into triangular (or quadrilateral) elements; number nodes and elements.
- Assume displacement functions u(x,y), v(x,y) in each element.
- Find [B], [D] and the element stiffness matrix [k].
- Assemble [K], apply loads and boundary conditions, solve [K]{U} = {F}.
- Back-calculate strains and stresses; plot the stress contours.

Rule: put a fine mesh at stress concentrations (holes, notches, corners); avoid very thin, long triangles.

2. Constant Strain Triangle (CST) Element

2.1 Description

- Three nodes (1, 2, 3) at the corners, numbered anticlockwise. Coordinates (x_i, y_i).
- Six dof per element: $\{q\} = \{u_1\ v_1\ u_2\ v_2\ u_3\ v_3\}^T$.



CST: 3 nodes, 6 dof

$u = a_1+a_2x+a_3y$

$v = a_4+a_5x+a_6y$

ε = constant over element

2.2 Displacement function

$$u = a_1 + a_2x + a_3y$$

$$v = a_4 + a_5x + a_6y$$

- The six constants are found from the six nodal displacements. For u:

$$u_1 = a_1 + a_2x_1 + a_3y_1$$

$$u_2 = a_1 + a_2x_2 + a_3y_2$$

$$u_3 = a_1 + a_2x_3 + a_3y_3$$

- Solving by Cramer's rule gives a_1, a_2, a_3 in terms of u_1, u_2, u_3

(similarly a_4, a_5, a_6 in terms of v's).

2.3 Shape functions

$$u = N_1u_1 + N_2u_2 + N_3u_3,\ v = N_1v_1 + N_2v_2 + N_3v_3$$

$$N_i = (\alpha_i + \beta_i x + \gamma_i y) / 2A,\ i = 1, 2, 3$$

$$\alpha_1 = x_2y_3 - x_3y_2,\ \beta_1 = y_2 - y_3,\ \gamma_1 = x_3 - x_2$$

$$\alpha_2 = x_3y_1 - x_1y_3,\ \beta_2 = y_3 - y_1,\ \gamma_2 = x_1 - x_3$$

$$\alpha_3 = x_1y_2 - x_2y_1,\ \beta_3 = y_1 - y_2,\ \gamma_3 = x_2 - x_1$$

Area of triangle: $2A = x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)$

- Check: $N_1 + N_2 + N_3 = 1$ for every point; in area coordinates $N_i = L_i$.
- In matrix form $\{u\ v\}^T = [N]\{q\}$ with $[N] = [N_1\ 0\ N_2\ 0\ N_3\ 0\ ;\ 0\ N_1\ 0\ N_2\ 0\ N_3]$.

3. Strain – Displacement Matrix of CST

$$\varepsilon_x = \partial u/\partial x = (\beta_1u_1 + \beta_2u_2 + \beta_3u_3) / 2A$$

$$\varepsilon_y = \partial v/\partial y = (\gamma_1v_1 + \gamma_2v_2 + \gamma_3v_3) / 2A$$

$$\gamma_{xy} = \partial u/\partial y + \partial v/\partial x = (\gamma_1u_1 + \beta_1v_1 + \gamma_2u_2 + \beta_2v_2 + \gamma_3u_3 + \beta_3v_3) / 2A$$

$$\{\varepsilon\} = [B]\{q\}$$

$$[B] = (1/2A)\ [\beta_1\ 0\ \beta_2\ 0\ \beta_3\ 0\ ;\ 0\ \gamma_1\ 0\ \gamma_2\ 0\ \gamma_3\ ;\ \gamma_1\ \beta_1\ \gamma_2\ \beta_2\ \gamma_3\ \beta_3]$$

- [B] contains only nodal coordinates (no x or y) – so strains are

CONSTANT throughout the element. That is why it is called the constant strain triangle.

- Because stress = [D][B]{q}, stress is also constant in the element; the value corresponds to the centroid.

4. Stress – Strain Relation and Stiffness Matrix of CST

4.1 Stress

$$\{\sigma\} = [D]\{\varepsilon\} = [D][B]\{q\}$$

Plane stress: $[D] = E/(1-\nu^2)\ [1\ \nu\ 0\ ;\ \nu\ 1\ 0\ ;\ 0\ 0\ (1-\nu)/2]$

Plane strain: $[D] = E/[(1+\nu)(1-2\nu)]\ [1-\nu\ \nu\ 0\ ;\ \nu\ 1-\nu\ 0\ ;\ 0\ 0\ (1-2\nu)/2]$

4.2 Element stiffness matrix

- From the minimum potential energy principle:

$$[k] = \int_V [B]^T [D][B]\, dV$$

- [B] and [D] are constants, so for an element of uniform thickness t:

$$[k] = t\, A\, [B]^T\, [D]\, [B]$$

- The matrix is 6×6, symmetric, and singular (rigid body modes), with positive diagonal terms.

4.3 Nodal load vectors

- Body force (self weight) per unit volume $\{X\ Y\}$:

  $$\{f_b\} = (t\,A/3)\,\{X\ Y\ X\ Y\ X\ Y\}^T$$

- Traction on an edge of length l_{12} (uniform pressure p_x, p_y):

  $$\{f_T\} = (t\,l_{12}/2)\,\{p_x\ p_y\ p_x\ p_y\ 0\ 0\}^T$$

- Initial strain / temperature: $\{f_0\} = t\,A\,[B]^T[D]\{\varepsilon_0\}$, $\{\varepsilon_0\} = \{\alpha\Delta T\ \alpha\Delta T\ 0\}^T$ (plane stress).

4.4 Properties of CST

- Simple and easy; fits irregular boundaries; constant strain – poor accuracy where stress changes quickly, so many small elements are required.
- Displacement is continuous (linear) along each edge – compatibility is satisfied.
- Convergence is slow compared with higher order elements.

Solved example outline: given nodal coordinates, find A, β_i, γ_i, $[B]$, then $[k] = tA[B]^T[D][B]$ and $\{\sigma\} = [D][B]\{q\}$.

5. Worked Example for CST

- Problem: Triangle with nodes 1 (0, 0), 2 (3, 0), 3 (0, 2) mm, $t = 1$ mm, $E = 200$ GPa, $\nu = 0.3$, plane stress. Find $[B]$ and the area.

  $$2A = 0(0-2) + 3(2-0) + 0(0-0) = 6 \rightarrow A = 3\ \text{mm}^2$$

  $$\beta_1 = y_2 - y_3 = -2,\ \beta_2 = y_3 - y_1 = 2,\ \beta_3 = y_1 - y_2 = 0$$

$\gamma_1 = x_3 - x_2 = -3,\ \gamma_2 = x_1 - x_3 = 0,\ \gamma_3 = x_2 - x_1 = 3$

$[B] = (1/6)\ [-2\ 0\ 2\ 0\ 0\ 0\ ;\ 0\ -3\ 0\ 0\ 0\ 3\ ;\ -3\ -2\ 0\ 2\ 3\ 0]$

- $[D]$ for plane stress: $E/(1-\nu^2) = 200000/0.91 = 2.198 \times 10^5$ N/mm^2

$[D] = 2.198\times10^5\ [1\ 0.3\ 0\ ;\ 0.3\ 1\ 0\ ;\ 0\ 0\ 0.35]$

- Suppose the nodal displacements are $u_2 = 0.001$ mm, all others zero. Then

$\varepsilon_x = \beta_2 u_2 / 2A = 2 \times 0.001 / 6 = 3.33 \times 10^{-4}$

$\varepsilon_y = 0,\ \gamma_{xy} = \gamma_2 u_2 / 2A = 0$ (since $\gamma_2 = 0$)

$\sigma_x = 2.198\times10^5 \times 3.33\times10^{-4} = 73.3$ N/mm^2, $\sigma_y = 0.3\,\sigma_x = 22$ N/mm^2, $\tau_{xy} = 0$

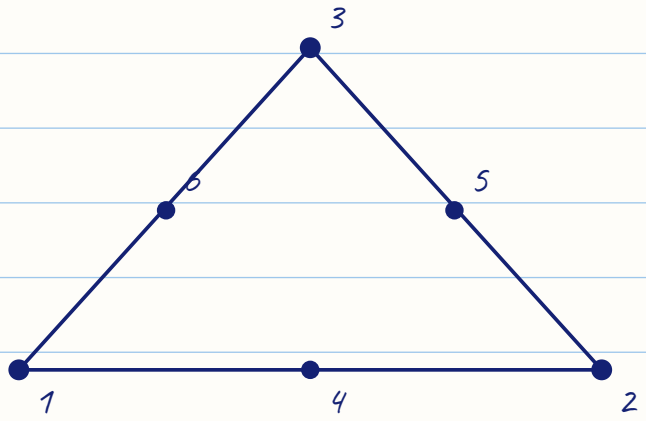
6. Linear Strain Triangle (LST) Element

6.1 Description

- Six nodes: three corner nodes and three mid-side nodes. 12 dof per element.
- Displacement is a complete quadratic polynomial:

$u = a_1 + a_2x + a_3y + a_4x^2 + a_5xy + a_6y^2$

$v = a_7 + a_8x + a_9y + a_{10}x^2 + a_{11}xy + a_{12}y^2$



LST: 6 nodes, 12 dof

$u = a_1 + a_2x + a_3y + a_4x^2 + a_5xy + a_6y^2$

strain varies linearly

6.2 Shape functions (area coordinates)

$$N_1 = L_1(2L_1 - 1),\ N_2 = L_2(2L_2 - 1),\ N_3 = L_3(2L_3 - 1)$$

$$N_4 = 4L_1L_2,\ N_5 = 4L_2L_3,\ N_6 = 4L_3L_1$$

- Sum of $N_i = 1$. Each $N_i = 1$ at its own node, 0 at others.

6.3 Strain and stress

- Strains are obtained by differentiating the shape functions; they vary LINEARLY within the element:

  $\varepsilon_x = \partial u/\partial x,\ \varepsilon_y = \partial v/\partial y,\ \gamma_{xy} = \partial u/\partial y + \partial v/\partial x$

  $[B]$ is a 3×12 matrix whose terms are linear functions of x and y.

  $[k] = t \int_A [B]^T [D] [B]\, dA$ (12 × 12)

- Integration is done using area-coordinate formula or Gauss numerical integration.
- Stress $\{\sigma\} = [D][B]\{q\}$ also varies linearly – more accurate than CST for the same number of nodes; curved boundaries can be fitted

with mid-side nodes.

6.4 CST vs LST

- CST: 3 nodes, 6 dof, constant strain, [k] 6×6, simple but needs fine mesh.
- LST: 6 nodes, 12 dof, linear strain, [k] 12×12, more accurate, more computation.
- LST gives good results with fewer elements; CST requires a large number of elements to reach the same accuracy.

7. Isoparametric Formulation

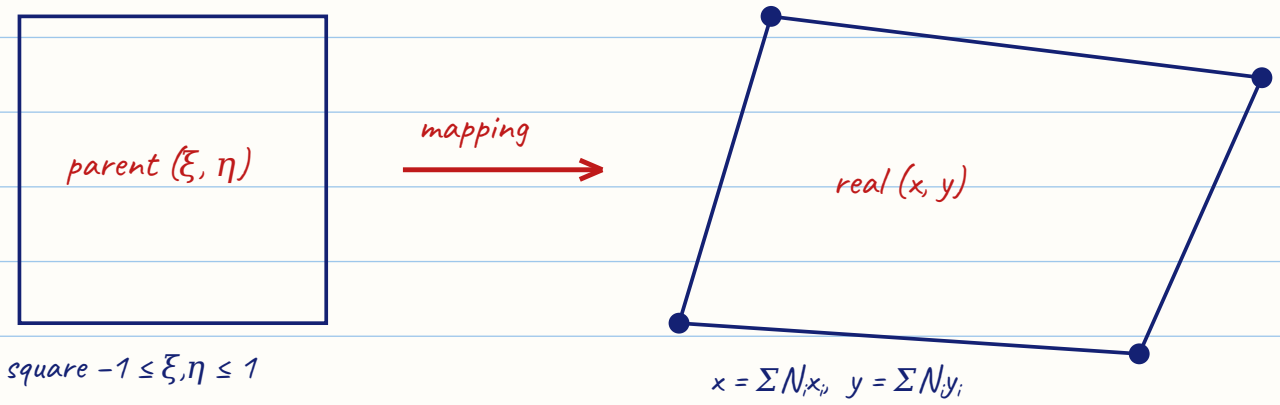
7.1 Concept

- Iso = same. In an isoparametric element, the SAME shape functions are used to define (i) the geometry (x, y) and (ii) the displacement (u, v).

$$x = \Sigma\, N_i(\xi,\eta)\, x_i,\ y = \Sigma\, N_i(\xi,\eta)\, y_i$$

$$u = \Sigma\, N_i(\xi,\eta)\, u_i,\ v = \Sigma\, N_i(\xi,\eta)\, v_i$$

- Subparametric: geometry uses lower-order functions than displacement. Superparametric: geometry uses higher order than displacement.
- A regular parent element in (ξ, η) is mapped to a distorted (curved or skew) element in (x, y).



7.2 Isoparametric quadrilateral (4 node)

$N_1 = \frac{1}{4}(1-\xi)(1-\eta)$, $N_2 = \frac{1}{4}(1+\xi)(1-\eta)$, $N_3 = \frac{1}{4}(1+\xi)(1+\eta)$, $N_4 = \frac{1}{4}(1-\xi)(1+\eta)$

7.3 Jacobian matrix

- Derivatives with respect to x, y are needed for strain, but N_i are functions of ξ, η. By chain rule:

$$\{\partial N/\partial\xi \ \partial N/\partial\eta\}^T = [J]\{\partial N/\partial x \ \partial N/\partial y\}^T$$

$$[J] = [\partial x/\partial\xi \ \partial y/\partial\xi \ ; \partial x/\partial\eta \ \partial y/\partial\eta] = [\Sigma(\partial N_i/\partial\xi)x_i \ \Sigma(\partial N_i/\partial\xi)y_i \ ; \Sigma(\partial N_i/\partial\eta)x_i \ \Sigma(\partial N_i/\partial\eta)y_i]$$

$$\{\partial N/\partial x \ \partial N/\partial y\}^T = [J]^{-1}\{\partial N/\partial\xi \ \partial N/\partial\eta\}^T$$

- Element area: $dA = dx\,dy = \det[J]\,d\xi\,d\eta$. $\det[J]$ must be positive everywhere – otherwise the element is too distorted (folded).

7.4 Strain–displacement matrix and stiffness

$\{\varepsilon\} = [B]\{q\}$, $[B]$ built from $\partial N_i/\partial x$ and $\partial N_i/\partial y$ (each node gives a 3×2 block)

$[B_i] = [\partial N_i/\partial x \; 0 \; ; \; 0 \; \partial N_i/\partial y \; ; \; \partial N_i/\partial y \; \partial N_i/\partial x]$

$[k] = t \int_{-1}^{1} \int_{-1}^{1} [B]^T [D] [B] \det[J] \, d\xi \, d\eta$

- The integrand is too complex for exact integration; Gauss quadrature is used (2×2 points for a 4-node element, 3×3 for 8 or 9 node).

7.5 Advantages of isoparametric elements

- Curved boundaries and irregular shapes handled accurately with one element.
- Same code for any shape; integration done in the fixed domain -1 to +1.
- Convergence criteria satisfied automatically (rigid body motion + constant strain), compatible along boundaries.
- Widely used in commercial software (ANSYS, ABAQUS, NASTRAN).

8. Isoparametric Elements for 2-D Analysis

- 4-node quadrilateral (Q4): bilinear, 8 dof.
- 8-node quadrilateral (Q8, serendipity): quadratic, curved edges, 16 dof.

- 9-node quadrilateral (Q9, Lagrange): 18 dof.
- Triangular isoparametric elements: 3-node and 6-node (area coordinates).
- Procedure summary:
- Step 1: Select the parent element and shape functions $N_i(\xi,\eta)$.
- Step 2: Compute $[J]$ and $\det[J]$ at the Gauss point.
- Step 3: Compute $[B]$ = function of $[J]^{-1}$ and $\partial N/\partial\xi$, $\partial N/\partial\eta$.
- Step 4: Evaluate $[B]^T[D][B]\det[J]$ at each Gauss point and sum with weights.

  $$[k] \approx t \Sigma_i \Sigma_j w_i w_j [B]^T[D][B] \det[J]$$
- Step 5: Assemble, apply boundary conditions, solve, compute stresses at Gauss points.

9. Formulation of CST Element – Summary

- Nodes 3, dof 6. Displacement linear: $u = a_1+a_2x+a_3y$.
- Shape functions $N_i = (\alpha_i + \beta_i x + \gamma_i y)/2A$.
- $[B]$ constant (3×6), strains constant.
- $[k] = t\,A\,[B]^T[D][B]$ (6×6).
- Stress $\{\sigma\} = [D][B]\{q\}$; load vectors from body force and edge traction.

Important questions: (1) derive $[B]$ and $[k]$ for CST, (2) compare CST and LST, (3) explain isoparametric formulation, (4) shape functions of 4-node and 8-node isoparametric elements, (5) a

numerical problem on CST.

1. Solution Techniques – Introduction

- After the elements are formed, we have to assemble them and solve a large set of simultaneous equations: $[K]\{U\} = \{F\}$.
- Steps involved: (i) numerical integration of element matrices, (ii) condensation of unwanted dof, (iii) assembly of elements, (iv) imposing boundary conditions and (v) solving the equations.
- For static loads the load does not change with time and inertia forces are ignored.

2. Numerical Integration

2.1 Need

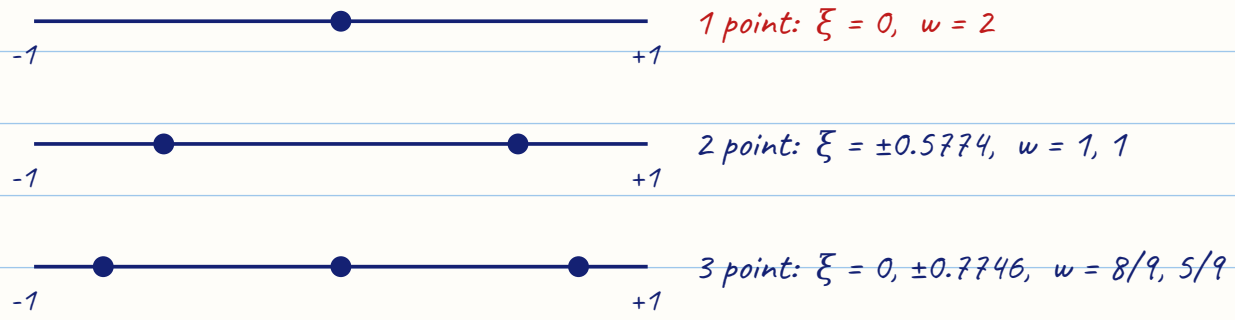
- In isoparametric elements the integrand $[B]^T[D][B]\det[J]$ is a complicated rational function of ξ and η – exact integration is not possible.
- So the integral is replaced by a weighted sum of the integrand values at selected sampling points.

$$\int_{-1}^{1} f(\xi)\, d\xi \approx \sum_{i=1}^{n} w_i f(\xi_i)$$

2.2 Newton–Cotes and Gauss quadrature

- Newton–Cotes (trapezoidal, Simpson): equally spaced points; n points integrate exactly a polynomial of degree (n-1).
- Gauss quadrature: sampling points and weights chosen optimally; n points integrate exactly a polynomial of degree (2n-1). Fewer points needed – hence preferred in FEM.

2.3 Gauss points and weights (limits -1 to +1)



- n = 1: ξ = 0, w = 2 (exact for linear polynomial).
- n = 2: ξ = ±1/√3 = ±0.57735, w = 1 each (exact for cubic).
- n = 3: ξ = 0, ±√(3/5) = ±0.77460, w = 8/9, 5/9, 5/9 (exact for degree 5).
- n = 4: ξ = ±0.33998, ±0.86114, w = 0.65215, 0.34785.

2.4 Change of limits

- If the integral is over [a, b], convert to the natural interval [-1, 1] by

$$x = (b - a)/2 \cdot \xi + (b + a)/2, \quad dx = (b - a)/2 \, d\xi$$

$$\int_a^b f(x)\,dx = (b - a)/2 \int_{-1}^{1} f(\xi)\,d\xi$$

2.5 Two-dimensional integration

$$\int_{-1}^{1}\int_{-1}^{1} f(\xi,\eta)\,d\xi\,d\eta \approx \sum_{i=1}^{n}\sum_{j=1}^{n} w_i\, w_j\, f(\xi_i, \eta_j)$$

- 2×2 Gauss rule (4 points): ξ, η = ±0.57735, weights 1×1 = 1.
- 3×3 Gauss rule (9 points): weights 25/81, 40/81, 64/81.

- Choice of order: Q4 element → 2×2; Q8 / Q9 element → 3×3. Too few points causes zero-energy (hourglass) modes; too many gives no extra accuracy.

2.6 Solved example

- Problem: Evaluate $I = \int_{-1}^{1} (3e^x + x^2 + 1/(x+2))\, dx$ by 2-point Gauss quadrature.

  Points: $x = \pm 0.57735$, $w = 1$.

  $f(0.57735) = 3(1.7813) + 0.3333 + 1/2.57735 = 5.3439 + 0.3333 + 0.3880 = 6.0652$

  $f(-0.57735) = 3(0.5614) + 0.3333 + 1/1.42265 = 1.6842 + 0.3333 + 0.7029 = 2.7204$

  $I \approx 1 \times 6.0652 + 1 \times 2.7204 = 8.7856$
- Exact value: $3(e - 1/e) + 2/3 + \ln 3 = 7.0512 + 0.6667 + 1.0986 = 8.8165$. The error is small (about 0.4%).

3. Static Condensation

3.1 Concept

- Also called substructuring or Guyan reduction. Some dof (usually internal / interior nodes) are eliminated at the element or substructure level, so the global system becomes smaller.
- Dof are divided into master (retained) $\{u_m\}$ and slave (condensed) $\{u_s\}$ dof.

K_{mm}	K_{ms}
K_{sm}	K_{ss}

$$\begin{Bmatrix} u_m \\ u_s \end{Bmatrix} = \begin{Bmatrix} F_m \\ F_s \end{Bmatrix}$$

m = master (retained)

s = slave (condensed)

$$K^* = K_{mm} - K_{ms}K_{ss}^{-1}K_{sm}$$

3.2 Derivation

$$[K_{mm}\ K_{ms}\ ;\ K_{sm}\ K_{ss}]\{u_m\ ;\ u_s\} = \{F_m\ ;\ F_s\}$$

- From the second row: $K_{sm}u_m + K_{ss}u_s = F_s \rightarrow u_s = K_{ss}^{-1}(F_s - K_{sm}u_m)$
- Substitute in the first row: $K_{mm}u_m + K_{ms}K_{ss}^{-1}(F_s - K_{sm}u_m) = F_m$

$$[K^*]\{u_m\} = \{F^*\}$$

$$[K^*] = [K_{mm}] - [K_{ms}][K_{ss}]^{-1}[K_{sm}]$$

$$\{F^*\} = \{F_m\} - [K_{ms}][K_{ss}]^{-1}\{F_s\}$$

- After solving for u_m, the slave displacements are recovered (back-substitution):

$$\{u_s\} = [K_{ss}]^{-1}\ (\{F_s\} - [K_{sm}]\{u_m\})$$

3.3 Advantages

- Reduces the size of the global matrix and the solution time; less storage.
- Useful in dynamic analysis (reducing dof) and for large repeated

substructures (e.g. aircraft wing, ship hull).

- No approximation is made in static analysis – the results for retained dof are exact (Guyan reduction is approximate only for dynamics).

3.4 Solved example

- Problem: Condense out dof 2 from $[K] = k\,[2\ -1\ 0;\ -1\ 2\ -1;\ 0\ -1\ 1]$ (dof 1, 2, 3), keeping dof 1 and 3.

  Master: 1, 3. Slave: 2. $K_{ss} = 2k$, $K_{ms} = k\,\{-1\ -1\}^T$, $K_{mm} = k\,[2\ 0;\ 0\ 1]$

  $[K^*] = k\,[2\ 0;\ 0\ 1] - (k^2/2k)\,[1\ 1;\ 1\ 1] = k\,[1.5\ -0.5;\ -0.5\ 0.5]$

- The reduced 2×2 matrix is symmetric and of reduced size.

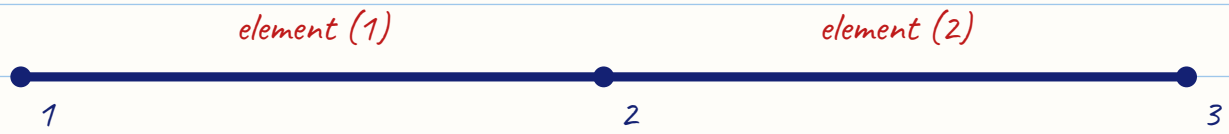
4. Assembly of Elements

4.1 Global stiffness matrix

- Individual element matrices are combined to form the global matrix by the direct stiffness method:

  $[K] = \Sigma_{e=1}^{n}\,[k^{(e)}]$, $\{F\} = \Sigma\,\{f^{(e)}\}$

- The element local dof are mapped onto the global dof using the connectivity (element–node table); each term k_{ij} of the element matrix is added to the global location K_{IJ}.



$[K^{(1)}]$ → rows/cols 1,2 $[K^{(2)}]$ → rows/cols 2,3

overlap at node 2: $K_{22} = k_{22}^{(1)} + k_{11}^{(2)}$

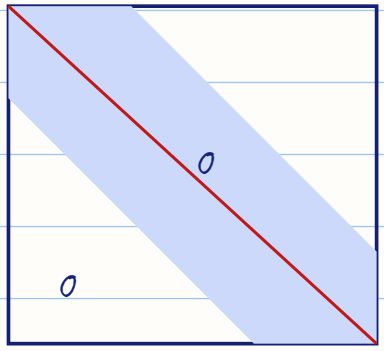
$[K] = \Sigma\ [k^{(e)}]$ (direct stiffness method)

4.2 Steps

- Step 1: Prepare the connectivity table (element, node i, node j) and the global dof numbering.
- Step 2: Initialise $[K] = 0$ of size (total dof)^2.
- Step 3: For each element, compute $[k^{(e)}]$ in the global coordinate system (transform if inclined).
- Step 4: Add $k_{ij}^{(e)}$ to K_{IJ} where I, J are the global dof numbers.
- Step 5: Similarly assemble the load vector.
- Terms at shared nodes are added, other terms are copied.

4.3 Properties of [K]

- Symmetric: $K_{ij} = K_{ji}$. Banded: non-zero terms near the diagonal, due to local connectivity of elements.
- Sparse: most terms are zero. Singular before boundary conditions are applied.
- Positive diagonal terms; positive definite after supports are imposed.



Banded, symmetric, sparse

Only non-zero band is stored

Half-bandwidth: nb = (D+1)·f

D = max node-number difference in an element

4.4 Bandwidth

Half bandwidth $nb = (D + 1)\, f$

- D = maximum difference between node numbers of any element, f = dof per node.
- Efficient numbering of nodes (to reduce D) reduces storage and computing time; number nodes along the shorter direction of the structure.
- Methods: banded storage, skyline (profile) storage, and frontal solution.

4.5 Example – assembly of 3 bars in series

- Nodes 1-2-3-4, elements (1-2), (2-3), (3-4) with stiffnesses k_1, k_2, k_3:

$$[K] = [\, k_1 \; -k_1 \; 0 \; 0 \; ; \; -k_1 \; k_1+k_2 \; -k_2 \; 0 \; ; \; 0 \; -k_2 \; k_2+k_3 \; -k_3 \; ; \; 0 \; 0 \; -k_3 \; k_3 \,]$$

- Half bandwidth for a bar: $nb = (1 + 1) \times 1 = 2$.

5. Boundary Conditions

5.1 Types

- Essential (geometric) boundary conditions: prescribed displacements ($u = 0$ at a fixed support). Must be applied.
- Natural (force) boundary conditions: prescribed loads / tractions – included in the force vector.

5.2 Elimination approach

- Remove the rows and columns of the fixed dof from $[K]$ and $\{F\}$; solve the remaining reduced system. Reactions are calculated afterwards:

  $$\{R\} = [K]\{U\} - \{F\}$$

- For a non-zero prescribed displacement a at dof j: subtract $K_{ij} \cdot a$ from the right side of every other equation i and then delete row j.

5.3 Penalty approach

- Add a very large number C to the diagonal K_{jj} and $C \cdot a$ to F_j.

  $$(K_{jj} + C)\, U_j + \Sigma\, K_{ji} U_i = F_j + C\, a \rightarrow U_j \approx a$$

- Size of the matrix is unchanged, so the programming is easy. Typical $C = 10^4$ to 10^6 times the largest diagonal term.

  Reaction at the support: $R_j = -C\,(U_j - a)$

5.4 Multi-point constraints (short note)

- Relations of the form $\beta_1 U_i + \beta_2 U_j = \beta_0$ (inclined rollers, rigid links) are handled by penalty or Lagrange multiplier methods.

6. Solution Techniques for Static Loads

6.1 Direct methods

- Gauss elimination: reduce $[K]$ to upper triangular form by forward elimination, then back-substitution.
- Cholesky (LDL^T) decomposition: for symmetric positive-definite $[K]$, $[K] = [L][L]^T$; solve $[L]\{y\} = \{F\}$, then $[L]^T\{U\} = \{y\}$. Takes advantage of symmetry and banded form; about half the effort of Gauss elimination.
- Frontal solver: assembles and eliminates equations at the same time, so only the active front is kept in memory.
- Used in most commercial software because they are reliable and give the exact answer (up to round-off).

6.2 Gauss elimination steps

- Step 1: Eliminate U_1 from equations 2, 3, ... n using equation 1:

  $K_{ij}^{(1)} = K_{ij} - (K_{i1}/K_{11})\, K_{1j}$, $F_i^{(1)} = F_i - (K_{i1}/K_{11})\, F_1$
- Step 2: Continue for U_2, U_3, ... to get an upper triangular system.
- Step 3: Back substitution:

  $U_n = F_n^*/K_{nn}^*$, $U_i = (F_i^* - \Sigma_{j>i} K_{ij}^* U_j) / K_{ii}^*$
- Pivot (diagonal) terms must be non-zero and positive for a stable

structure; a zero pivot signals a mechanism (insufficient supports).

6.3 Iterative methods

- Gauss–Seidel, Jacobi, conjugate gradient. Start with an assumed {U} and improve it step by step until the change is small.

$$U_i^{(k+1)} = (1/K_{ii}) \left(F_i - \Sigma_{j<i} K_{ij} U_j^{(k+1)} - \Sigma_{j>i} K_{ij} U_j^{(k)} \right)$$

- Good for very large 3-D problems because only non-zero terms are stored; convergence depends on conditioning of [K].

6.4 Direct vs iterative

- Direct: fixed number of operations, robust, good for many load cases, needs more memory.
- Iterative: less memory, suited to very large sparse systems, convergence not guaranteed.

7. Complete Solution Flow (Static Analysis)

- (1) Pre-processing: geometry, mesh, material properties, loads, supports.
- (2) Element matrices: $[k^{(e)}]$ using shape functions; numerical integration if needed; condense internal dof.
- (3) Assembly: $[K]\{U\} = \{F\}$ using connectivity.
- (4) Apply boundary conditions (elimination or penalty).
- (5) Solve using Gauss elimination / Cholesky / iterative solver.
- (6) Post-processing: displacements, strains $\{\varepsilon\} = [B]\{q\}$, stresses

$\{\sigma\} = [D]\{\varepsilon\}$, reactions; check equilibrium.

- Mesh convergence: refine the mesh until the results (stress, displacement) no longer change significantly (h-refinement), or increase the polynomial order (p-refinement).

8. Solved Example – Assembly, Boundary Conditions and Soluti

- Problem: Two bars in series (nodes 1-2-3), each with AE/l = 1000 N/mm. Node 1 is fixed; loads: 500 N at node 2 and 300 N at node 3. Find the displacements.

  $[K] = 1000\ [1\ -1\ 0\ ;\ -1\ 2\ -1\ ;\ 0\ -1\ 1]$

  After $U_1 = 0$: $1000\ [2\ -1\ ;\ -1\ 1]\ \{U_2\ U_3\}^T = \{500\ 300\}^T$

  Gauss elimination: $R_2 \leftarrow R_2 + 0.5\ R_1 \rightarrow 1000\ (0.5)\ U_3 = 300 + 0.5 \times 500 = 550$

  $U_3 = 550/500 = 1.1$ mm

  $2000\ U_2 - 1000 \times 1.1 = 500 \rightarrow U_2 = 0.8$ mm

- Reaction at node 1: $R_1 = -1000\ (U_2) = -800$ N (balances the total load 500 + 300 = 800 N□).
- Bar forces: $P_1 = 1000(U_2 - U_1) = 800$ N, $P_2 = 1000(U_3 - U_2) = 300$ N.

9. Quick Revision Points

- Gauss quadrature with n points is exact up to polynomial degree 2n-1; weights: 1 point (2), 2 points (1, 1).
- Static condensation: $K^* = K_{mm} - K_{ms}K_{ss}^{-1}K_{sm}$.

- Assembly: add the element terms at common dof; [K] is symmetric, banded and sparse.
- Boundary conditions: elimination or penalty method.
- Solution: Gauss elimination / Cholesky for direct, Gauss–Seidel for iterative.

Important questions: (1) explain Gauss quadrature with example, (2) derive static condensation, (3) assembly of global stiffness matrix with banded form, (4) elimination and penalty approaches, (5) solve a small system by Gauss elimination.